

Real-Time Retail Feedback Intelligence

— A GenAI-powered Customer Review Automation Solution —



ChicStyle — a growing women's fashion retail platform

- ChicStyle sees massive spikes in customer activity during festive and holiday sales, leading to a surge in incoming reviews.
- Reviews arrive continuously and vary widely—from positive feedback to urgent issues like fit problems, delivery delays, defects, and sizing concerns.
- Delays in responding during these high-pressure periods can frustrate customers, damage trust, and negatively impact both immediate sales and long-term loyalty.
- To maintain customer satisfaction and protect brand reputation, ChicStyle needs a system that can process large volumes of reviews quickly, accurately, and with business context.



Statement:

- Retailers need an intelligent feedback system that can process high volumes of real-time reviews, detect sentiment, identify relevant products or services, and flag urgent issues.
- Traditional NLP models struggle due to **rule-based approaches and limited training data**, making them less effective for complex feedback.
- In contrast, generative AI understands context, distinguishes multiple opinions, and detects nuanced emotions, enabling more accurate insights and faster responses.

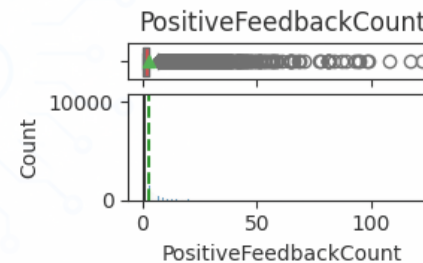
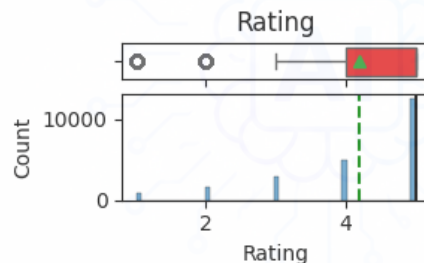
Objectives:

- Analyze and categorize feedback with real-time sentiment detection
- Identify the product/service referenced in each review
- Summarize insights by category and urgency

Impact: Enables faster issue resolution, improves product quality, and converts unstructured feedback into real-time business insights



- ▶ **Title:** The title of the review.
- ▶ **Review.Text:** The main body of the customer's review text.
- ▶ **Rating:** The product score given by the customer, from 1 (Worst) to 5 (Best).
- ▶ **Recommended:** A binary variable indicating if the customer recommends the product (1 for recommended, 0 for not recommended).
- ▶ **Positive Feedback Count:** The number of other customers who found the review helpful (Positive Integer).



A correlation heatmap titled "Correlation HeatMap" showing the relationships between four variables: Age, Rating, PositiveFeedbackCount, and Recommendation. The diagonal elements are all 1.0, indicating perfect self-correlation. The off-diagonal elements show the correlation between different variables: Age and Rating (0.0), Age and PositiveFeedbackCount (0.0), Age and Recommendation (0.0), Rating and PositiveFeedbackCount (-0.1), Rating and Recommendation (0.8), and PositiveFeedbackCount and Recommendation (-0.1). The color scale ranges from 0.0 (dark red) to 1.0 (dark blue).

	Age	Rating	PositiveFeedbackCount	Recommendation
Age	1.0	0.0	0.0	0.0
Rating	0.0	1.0	-0.1	0.8
PositiveFeedbackCount	0.0	-0.1	1.0	-0.1
Recommendation	0.0	0.8	-0.1	1.0

Figure 10 displays four word clouds comparing high and low ratings, categorized by custom stopwords. The top row shows the full word clouds for 'High Rating' and 'Low Rating'. The bottom row shows the word clouds for 'High Rating - custom stopwords' and 'Low Rating - custom stopwords', where stopwords have been removed. The word clouds for high ratings are dominated by terms like 'dress', 'love', 'top', 'size', 'color', 'wear', 'look', 'perfect', 'fabric', 'material', 'even', 'way', 'made', 'back', 'really', 'model', 'small', 'one', 'ordered', 'love', 'wear', 'shirt', 'looked', 'really', 'much', 'back', 'color', 'size', 'dress', 'look', 'fabric'. The word clouds for low ratings are dominated by terms like 'great', 'love', 'comfortable', 'perfect', 'way', 'really', 'back', 'retailer', 'model', 'even', 'large', 'small', 'sweater', 'work', 'made', 'cute', 'really', 'back', 'love', 'way', 'great', 'retailer', 'model', 'even', 'large'.



- **Customer Ratings:** Highly skewed toward positive, with **75%+ rated 5**, but lower ratings carry more informational value.
- **Key Drivers:** **Fit/size and appearance (color/style)** are the main factors influencing satisfaction—especially for Dresses and Tops.
- **Category Performance:**
 - Strong: Intimate, Bottoms, Jackets (functionality-driven)
 - Weak: Dresses, Tops, Trend (appearance-driven expectations)
- **Customer Behavior Insight:** Reviews with low ratings and no recommendations are perceived as more helpful, indicating higher decision impact.
- **Modeling Implication:** Presence of **mixed sentiment within single reviews** highlights the need for advanced AI (e.g., generative models) to capture nuanced feedback accurately.



Review Text is the key input for the LLM

- Ignore missing reviews (low %; no response expected) 
(used in this project)
- Alternative: use default input ("No comment") via prompt logic

LLM advantages:

- Handles misspellings & incomplete sentences
- Minimal preprocessing needed

Data handling:

- Missing categorical fields (Division, Department, ClassName) → excluded
- No preprocessing needed for Rating, Recommendation, PositiveFeedbackCount (no missing values)



- Zero-Shot prompt
 - Define model role and required outputs
 - Enforce strict JSON format
 - Specify allowed labels
 - No examples provided
- Few-Shot prompt
 - Add 3–5 examples before the task
 - Improves consistency for tricky cases: Fit vs Quality, mixed sentiment, primary vs secondary issues
 - Includes field-specific rules
- Chain-of-Thought prompt
 - Uses structured internal reasoning
 - Does not output reasoning steps
 - Outputs final JSON only
 - Improves quality through step-by-step decision logic

Each prompt level builds on the previous one.



• Why RAG

- LLMs rely only on prompt input → can miss business context.
- Improves consistency across similar customer issues and feedbacks.
- Reduces hallucination and unsupported responses.
- Keeping the instructions clear and structured.
- Aligns outputs with real-world patterns and expectations.

• How RAG Improves Prompting Performance

- Retrieves top-k similar past cases before generation.
- Injects grounded context into the prompt.
- Guides the model toward more accurate classification and responses.
- Enhances CoT reasoning with real examples.

• Result

- Higher accuracy
- Better consistency
- More reliable, production-ready outputs

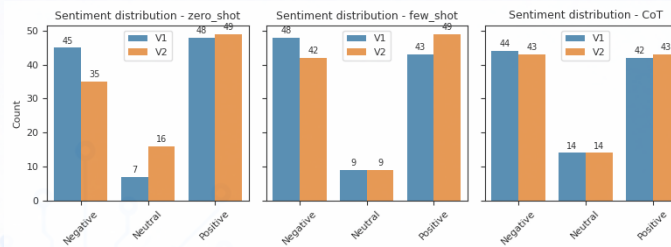
Note: In the proof-concept experiment, we use manually refined predictions (via ChatGPT) as ground truth for 20 reviews. The LLM judge compares outputs to these references based semantically. Although a small and curated dataset, it demonstrates clear improvements across metrics and overall performance.



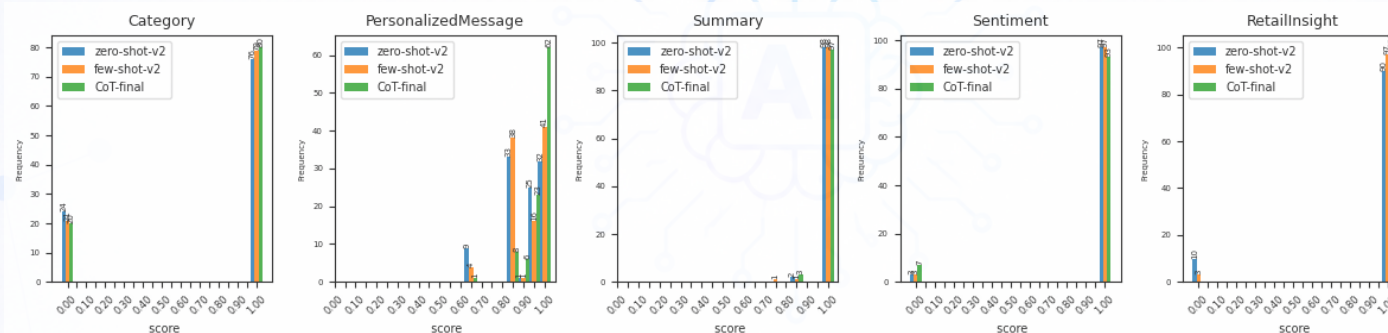
• Prompt performance average scores

Prompt	V1	V2	Final
RAG+CoT	x	x	0.979
CoT	0.784	0.932	0.939
Few-Shot	0.780	0.929	x
Zero-Shot	0.772	0.906	x

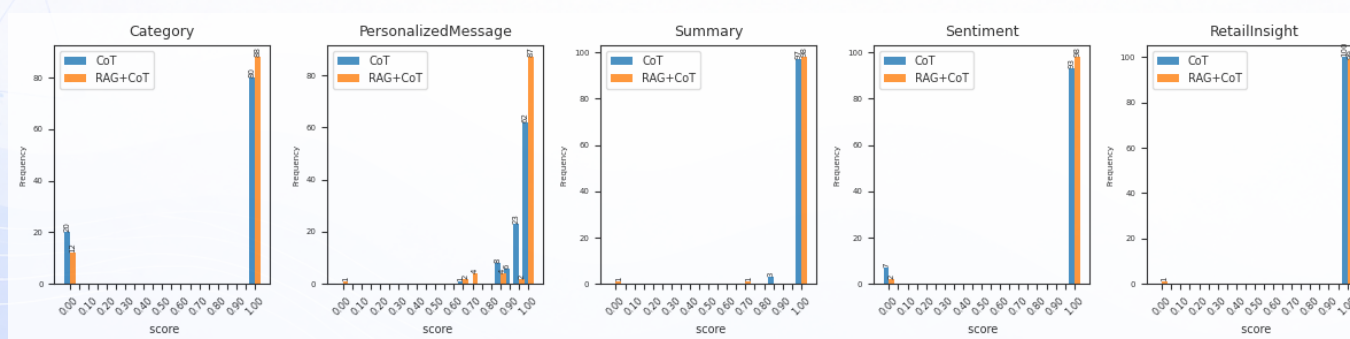
• Sentiment Prediction distribution



• LLM-judge score distributions by output field (Zero-Shot/Few-Shot/CoT)



• LLM-judge score distributions by output field (CoT/RAG+CoT)



Findings

- V2 > V1 across all techniques (better structured rules)
- Few-shot >> Zero-shot; CoT > Few-shot
- Scores (Category, Summary, Retail Insight) concentrated at high end
- CoT shows stable sentiment (less prompt drift)
- RAG + CoT achieves highest performance

Insights

- Best overall: **RAG + CoT**, followed by CoT
- CoT improves reasoning (infer → compare → decide → explain), reducing errors
- RAG adds grounding → less hallucination, more consistency
- RAG + CoT = most reliable and consistent results



● CoT Review Intelligence Engine

- Improves accuracy for mixed sentiment, fit, quality, delivery, urgency, and responses

● RAG-Grounded Response Layer

- Uses past cases + product data for grounded, consistent responses
- Reduces hallucination; improves traceability

● Agentic Framework + Harness

- Orchestrated LLM agents with automated evaluation for quality and continuous improvement

● Real-Time Dashboard

- Urgent issues, top problems, category insights
- Response status & sentiment trends

● Human-in-the-Loop

- Escalates high-urgency reviews to support teams



● Short-Term (3–6 Months)

- Deploy CoT-based classification (sentiment, category, urgency, summary) → faster, more accurate triage
- Add RAG-grounded responses (historical + product data) → consistent, brand-safe communication
- Launch real-time dashboard (sentiment, urgency, issues) → quicker decisions

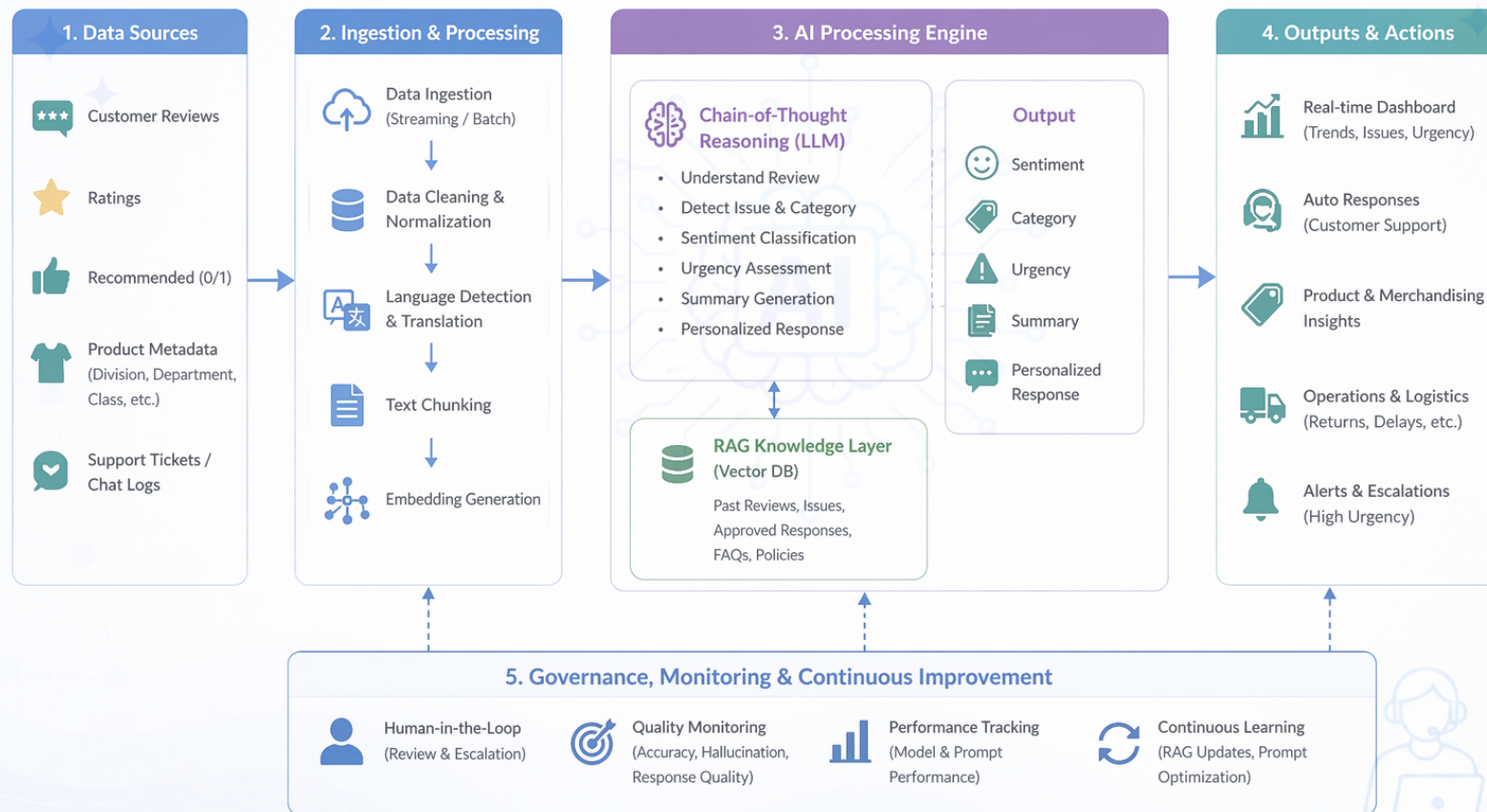
● Long-Term (6–12 Months)

- Integrate with CRM & auto-route high-urgency issues → faster resolution
- Build product trend detection (fit, fabric, sizing, delivery) → reduce returns
- Establish continuous learning & governance → ongoing improvement & risk reduction



ChicStyle – Generative AI Feedback Intelligence Platform

Simplified End-to-End Architecture



Estimated Benefits (Pilot & Early Scale)

- 30–50% less manual review effort
- 2–3× faster detection of negative/urgent issues
- More consistent categorization & summaries
- Earlier detection of product issues (fit, quality, delivery)
- Better customer experience with faster responses

Basis of Estimates

- 5–10 min manual review → seconds with CoT
- 10K–100K reviews/month
- Based on typical support workflows (excludes future RAG/CRM gains)

Structure (Pilot Phase)

Development Costs (One-time): ~\$6K–14K

Monthly Ops: ~\$400–1,000

- LLM API: \$200–500
- Cloud: \$100–300
- Maintenance: \$100–200

Cost per Review: ~\$0.002–0.01 → highly scalable



• Model Misclassification and Hallucination Risk

- Incorrect sentiment and urgency → wrong action
- Hallucinated summary and personalized message.
- Mitigation:
 - ✓ Model temperature control.
 - ✓ RAG implementation to ground answers based on historical curated responses.
 - ✓ human-in-the-loop for high urgency cases

• Cost Scalability

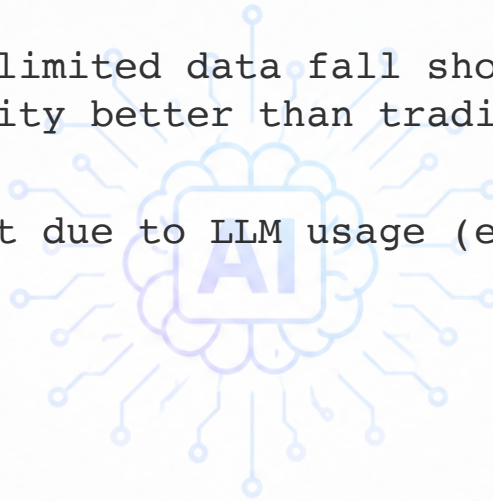
- High API usage during peak seasons
- Mitigation: hybrid prompt strategy

• Integration Complexity

- Connecting with CRM, support, product systems
- Mitigation: phased rollout



- When fixed rules and limited data fall short, GenAI handles complexity and ambiguity better than traditional ML
- **Trade-off:** Higher cost due to LLM usage (e.g., API quotas and service fees)





THANK YOU

Any Questions?

